

Create two instances in OCI as the backend set's (instance's) to be configured for the load balancer

Creating and Configuring Load Balancer In OCI

- 1) Login to OCI Console
- 2) Create a Virtual Cloud Network (VCN)
 - Navigate to Networking —> Virtual Cloud Network —> Create VCN
 - **Name:** Enter a name for your VCN, e.g., **myVCN**.
 - **Compartment:** Select the compartment for your VCN.
 - **CIDR Block:** Choose a CIDR block for your VCN, for example, **10.0.0.0/16**.
 - **DNS Resolution:** Keep it enabled.
 - **DNS Hostnames:** Keep it enabled.
 - Keep all other option default and Click **Create**.

The screenshot shows the 'Create a Virtual Cloud Network' form in the Oracle Cloud Console. The form includes the following sections:

- Name:** A text input field.
- Create in Compartment:** A dropdown menu showing 'pilot (root)'.
- IPv4 CIDR Blocks:** A section with a note: 'You can assign up to 5 IPv4 CIDR blocks to a VCN. There must be at least one IPv4 CIDR block assigned to a VCN. [Learn more](#).' Below this is a dropdown menu for selecting a CIDR block, with a hint 'See Example: 10.0.0.0/16'.
- Use DNS hostnames in this VCN:** A toggle switch that is currently turned on.
- DNS Label:** A text input field with the value 'Generated from virtual cloud network name if not specified'.
- DNS Domain Name:** A text input field with the value 'Generated from virtual cloud network name if not specified'.
- IPv6 Prefixes:** A section at the bottom of the form.

At the bottom of the form, there are three buttons: 'Create VCN', 'Save as stack', and 'Cancel'.

3) Create Subnets in the VCN

- After the VCN is created, navigate to the **Subnets** section inside VCN
- Click **Create Subnet**.
- **Name:** Enter a name, e.g., **subnet-AD1**.
- **Compartment:** Select the compartment.
- **CIDR Block:** Enter a subnet CIDR block. For example, **10.0.1.0/24** for the first subnet.
- **Availability Domain (AD):** Choose the **first AD** (e.g., **AD-1**).
- **Subnet Type:** Select **Public**.
- Keep all other option default and Click **Create**.

Now, create the second subnet:

- Click **Create Subnet** again.
- **Name:** **subnet-AD2**.
- **CIDR Block:** Enter another subnet CIDR block, e.g., **10.0.2.0/24**.
- **Availability Domain (AD):** Choose the **second AD** (e.g., **AD-2**).
- **Subnet Type:** Select **Public**.
- Click **Create**.

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Networking Virtual Cloud Networks Virtual Cloud Network Details

my-vcn

Move resource Add security attributes Add tags Delete

Details Security Tags

VCN Information

Compartment: ptpl (root)
 Created: Tue, Dec 10, 2024, 06:40:46 UTC
 IPv4 CIDR Block: 10.0.0.0/16
 IPv6 Prefix: -
 OCID: ...xw3ygmq Show Copy
 DNS Resolver: my-vcn
 Default Route Table: Default Route Table for my-vcn
 DNS Domain Name: myvcn.oraclevcn.com

Resources

Subnets (2)
 CIDR Blocks/Prefixes (1)
 Route Tables (1)
 Internet Gateways (1)
 Dynamic Routing Gateways Attachments (0)

Subnets

Create Subnet

Name	State	IPv4 CIDR Block	IPv6 Prefixes	Subnet Access	Created
Subnet-AD2	Available	10.0.2.0/24	---	Public (yFYg:AP-HYDERABAD-1-AD-1)	Tue, Dec 10, 2024, 06:55:05 UTC
Subnet-AD1	Available	10.0.1.0/24	---	Public (yFYg:AP-HYDERABAD-1-AD-1)	Tue, Dec 10, 2024, 06:54:42 UTC

Showing 2 items < 1 of 1

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Create Subnet

Name

Create in Compartment

ptpl (root)

Subnet Type

Regional (Recommended)
 Instances in the subnet can be created in any availability domain in the region. Useful for high availability.

Availability Domain-specific
 Instances in the subnet can only be created in one availability domain in the region.

Availability Domain

yFYg:AP-HYDERABAD-1-AD-1

IPv4 CIDR Block

IPv4 CIDR Block

Example: 10.0.0.0/16

IPv6 Prefixes

Maximum amount of 8 IPv6 prefixes per Subnet. IP ranges of the IPv6 prefixes must not overlap. [Learn More](#)

Route Tables in page (new) [Click here to create route table](#)
 Default Route Table for my-vcn

Subnet Access

Private Subnet
 Provide public IP addresses for instances in this Subnet.

Public Subnet
 Allow public IP addresses for instances in this Subnet.

Subnet Labels

Use Only (Only) Instances in this Subnet
 These subnets cannot be used for launching an instance.

Subnet Labels

Comment from subnet name if not specified
 This field can contain labels and other 10 characters max.

Subnet Domain Name - must only
 (Only Subnet Name is allowed here)

Drop Options in page (new) [Click here to create drop options](#)
 Loading...

Security Lists

This subnet is associated with 0 network security lists with the subnet.
 Security List in page (new) [Click here to create security list](#)

Click here to create security list

Create Subnet Cancel

4) Create an Internet Gateway (IGW)

- In the Oracle Cloud Console, under Networking, —> VCN —> My VCN —> click Internet Gateways.
- Click Create Internet Gateway.
- Name: Enter a name, e.g., myIGW.
- Compartment: Choose the correct compartment.
- Attach to VCN: Select the VCN you just created (myVCN).
- Keep all other option default and Click Create.

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Networking Virtual Cloud Networks Virtual Cloud Network Details Internet Gateways

my-vcn

Move resource Add security attributes Add tags Delete

Details Security Tags

VCN Information

Compartment: ptpl (root)
 Created: Tue, Dec 10, 2024, 06:40:46 UTC
 IPv4 CIDR Block: 10.0.0.0/16
 IPv6 Prefix: -
 OCID: ...xw3ygmq Show Copy
 DNS Resolver: my-vcn
 Default Route Table: Default Route Table for my-vcn
 DNS Domain Name: myvcn.oraclevcn.com

Resources

Subnets (2)
 CIDR Blocks/Prefixes (1)
 Route Tables (1)
 Internet Gateways (1)
 Dynamic Routing Gateways Attachments (0)

Internet Gateways

Create Internet Gateway

Name	State	Route Table	Created
my-vcn-internetGateway	Available	---	Tue, Dec 10, 2024, 06:43:46 UTC

Showing 1 item < 1 of 1

5) Update Route Tables

- After creating the IGW, navigate to the Route Tables section under Networking.
- Click on the default route table.

Add Route Rule:

- Destination CIDR Block: Enter **0.0.0.0/0** to allow all traffic to the internet.
- Target Type: Choose Internet Gateway.
- Target: Select the internet gateway (**myIGW**).
- Click Save Changes.

Oracle Cloud console showing the 'my-vcn' Virtual Cloud Network details. The 'Route Tables' section is highlighted, showing a table with one entry: 'Default Route Table for my-vcn' in an 'Available' state with 1 rule.

Oracle Cloud console showing the 'Default Route Table for my-vcn' details. The 'Route Rules' section is highlighted, showing a table with one entry: '0.0.0.0/0' with 'Internet Gateway' as the target type and 'my-vcn-internetGateway' as the target.

Oracle Cloud console showing the 'Add Route Rules' dialog box. The 'Route Rule' form is filled out with 'Internet Gateway' as the target type, '0.0.0.0/0' as the destination CIDR block, and 'my-vcn-internetGateway' as the target internet gateway.

6) Create a Load Balancer (LB)

- Navigate to Load Balancers
- In the Oracle Cloud Console, go to Networking and click Load Balancers.
- Click Create Load Balancer.
- Name: Enter a name for your Load Balancer, e.g., **myLoadBalancer**
- Compartment: Choose the compartment.
- Availability Domain: Select the first AD where the LB will be created (e.g., AD-1).
- VCN: Select the VCN you created earlier (myVCN).
- Subnet: Select the first public subnet (subnet-AD1).

The screenshot shows the 'Create load balancer' page in the Oracle Cloud Console. The page is divided into several sections for configuration:

- Name:** myLoadBalancer
- Compartment:** (selected)
- Availability Domain:** (selected)
- VCN:** (selected)
- Subnet:** (selected)
- Health Check:** (selected)
- Security:** (selected)
- Tagging:** (selected)

Configure Backend Set

- Name: Enter a name for the backend set, e.g., **backend-set-1**.
- Protocol: Choose the protocol to be used (e.g., HTTP or HTTPS).
- Port: Specify the backend port (e.g., **80** for HTTP).
- Health Check: Set up the health check:
- Type: Select HTTP.
- Port: Enter **80** (or the port you configured your backend instances to listen on).
- URL Path: Use **/health** or **/** (depending on how your backend instances respond to health checks).
- Interval: Set the interval (e.g., **30 seconds**).
- Keep all other option default and Click **Create**.

The screenshot shows the 'Configure Backend Set' section of the 'Create load balancer' page. The page is divided into several sections for configuration:

- Select backend servers:** (selected)
- Specify health check policy:** (selected)
- Backend set:** (selected)
- Max backend connections:** (selected)
- Security List:** (selected)

Configure Listener

- **Listener Name:** Enter a name, e.g., **http-listener**.
- **Protocol:** Choose HTTP (or HTTPS if using SSL).
- **Port:** Enter **80** (or **443** for HTTPS).
- **Backend Set:** Choose the backend set you created earlier (**backend-set-1**). (If There)
- **SSL certificate configurations** (update as required)

The screenshot shows the 'Create load balancer' wizard in the Oracle Cloud console, specifically the 'Configure listener' step. The left sidebar contains a progress bar with steps: 1. Add details, 2. Choose backends, 3. Configure listener (active), 4. Manage logging, and 5. Review and create. The main content area includes a description of a listener, a 'Listener name' field with the value 'listener_lb_2024-1216-1543', a 'Specify the type of traffic your listener handles' dropdown set to 'HTTPS', and a 'Specify the port your listener monitors for ingress traffic' dropdown set to '443'. Below these is the 'SSL certificate' section, which has a 'Certificate resource' dropdown set to 'Load balancer managed certificate'. Under 'Specify SSL certificate', the 'Choose SSL certificate file' radio button is selected. The 'SSL certificate' field is empty, with a prompt to 'Drop a file or select one'. There is also a 'Specify CA certificate' section with a toggle switch set to 'On' and options to 'Choose a CA file' or 'Paste a CA certificate'. At the bottom, there are 'Previous', 'Next', and 'Cancel' buttons. The footer includes 'Terms of Use and Privacy', 'Cookie Preferences', and a copyright notice for 2024.

Error Logs

- **Set the configurations as required or keep it default and click next**

The screenshot shows the 'Create load balancer' wizard in the Oracle Cloud console, specifically the 'Manage logging' step. The left sidebar shows the progress bar with steps: 1. Add details, 2. Choose backends, 3. Configure listener, 4. Manage logging (active), and 5. Review and create. The main content area is titled 'Error logs' and includes a description: 'Enabling access and error logs is optional, but recommended. Reviewing these logs can help you with diagnosing and fixing issues with your backend servers. Learn more about load balancer logging.' Below this is a toggle switch for 'Enabled' which is turned on. A note states: 'The error log captures detailed information about requests related to troubleshooting and monitoring. Learn more about error logs.' There are three fields: 'Compartment' set to 'ptpl (root)', 'Log name' set to 'lb_2024-1216-1543_error', and 'Log retention' set to 'One month (default)'. Below these is the 'Access logs' section, which has a toggle switch for 'Not enabled' which is turned off. A note states: 'The access log captures detailed information about requests sent to the load balancer. Learn more about access logs.' At the bottom, there are 'Previous', 'Next', and 'Cancel' buttons. The footer includes 'Terms of Use and Privacy', 'Cookie Preferences', and a copyright notice for 2024.

Review and Create

The screenshot shows the 'Create load balancer' wizard in the Oracle Cloud console, specifically the 'Review and create' step. The left sidebar shows the progress bar with steps: 1. Add details, 2. Choose backends, 3. Configure listener, 4. Manage logging, and 5. Review and create (active). The main content area is divided into two sections: 'Details' and 'Backends'. The 'Details' section includes fields for 'Load balancer name' (lb_2024-1216-1543), 'Visibility type' (Public), 'IP address' (Ephemeral IP address), 'Bandwidth' (Maximum: 10, Minimum: 10), 'Networking' (IPv6 address assignment: Disabled, VCN: my-vcn, Subnet: Subnet-AD1), 'Advanced options' (Management compartment: ptpl (root)), and 'Health check policy'. The 'Backends' section includes 'Load balancer policy' (Weighted round robin), 'Backend compartment' (ptpl (root)), and 'Backends' (No backends selected). At the bottom, there are 'Previous', 'Next', 'Submit', and 'Cancel' buttons. The footer includes 'Terms of Use and Privacy', 'Cookie Preferences', and a copyright notice for 2024.



ACTIVE

CMS-TnD

- Update shape
- Move resource
- Add security attributes
- Add tags
- Delete
- Create path analysis ▾

- Details
- Security
- Tags

Load balancer health

Overall health:  OK

Backend sets health

Critical:  0

Warning:  0